# Supplementary Information for Branch-specific detection of phylogenetic signal with SatuTe

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## Supplementary Results

### Baseline methodological controls

The baseline simulations remain necessary controls for the extension. They establish the gold-standard power curve, expose inflation from selecting and testing a topology on the same alignment, show the benefit and possible conservatism of taxon-pair Bonferroni correction, and document the distinct effect of substitution-model misspecification. The newly generated JC, GTR, protein and misspecification panels below reproduce these methodological comparisons using the current implementation and verified publication grids; no artwork from the earlier paper is reused.

### Topology-selection and bootstrap controls

| Model | Sites | Branch $t$ | Counterfactual | Refined | Difference |
|---|---|---|---|---|---|
| $+I+G4$ | 100 | 50 | 0.100 | 0.115 | 0.015 |
| | 500 | 40 | 0.342 | 0.384 | 0.042 |
| | 1,000 | 45 | 0.406 | 0.456 | 0.050 |
| | 5,000 | 60 | 0.482 | 0.584 | 0.102 |
| | 10,000 | 60 | 0.718 | 0.835 | 0.116 |
| $+I+R4$ | 100 | 10 | 0.558 | 0.573 | 0.015 |
| | 500 | 20 | 0.405 | 0.450 | 0.044 |
| | 1,000 | 22.5 | 0.449 | 0.522 | 0.073 |
| | 5,000 | 30 | 0.501 | 0.605 | 0.104 |
| | 10,000 | 35 | 0.402 | 0.565 | 0.163 |

Table S1: **Largest calibrated power gain at each alignment length in the invariant-mixture grid.** Differences are refined minus counterfactual; both formulas evaluate every alignment.

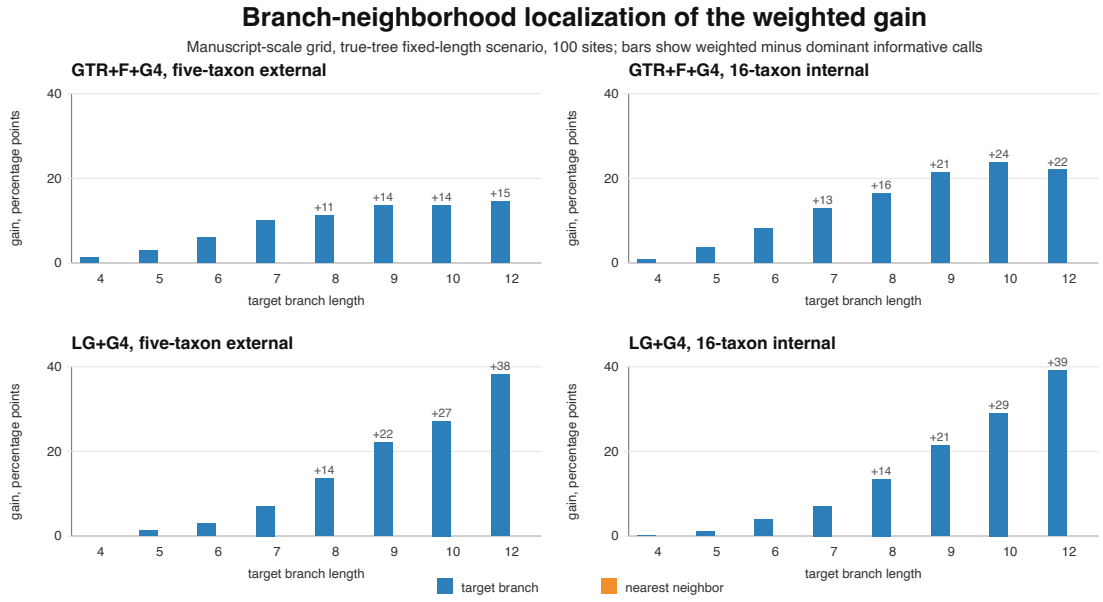



Figure S1: **Branch-neighbourhood localization.** Weighted-minus-dominant changes in informative-call fraction for the tested branch and adjacent branches. The change is concentrated on the deliberately lengthened focal branch.

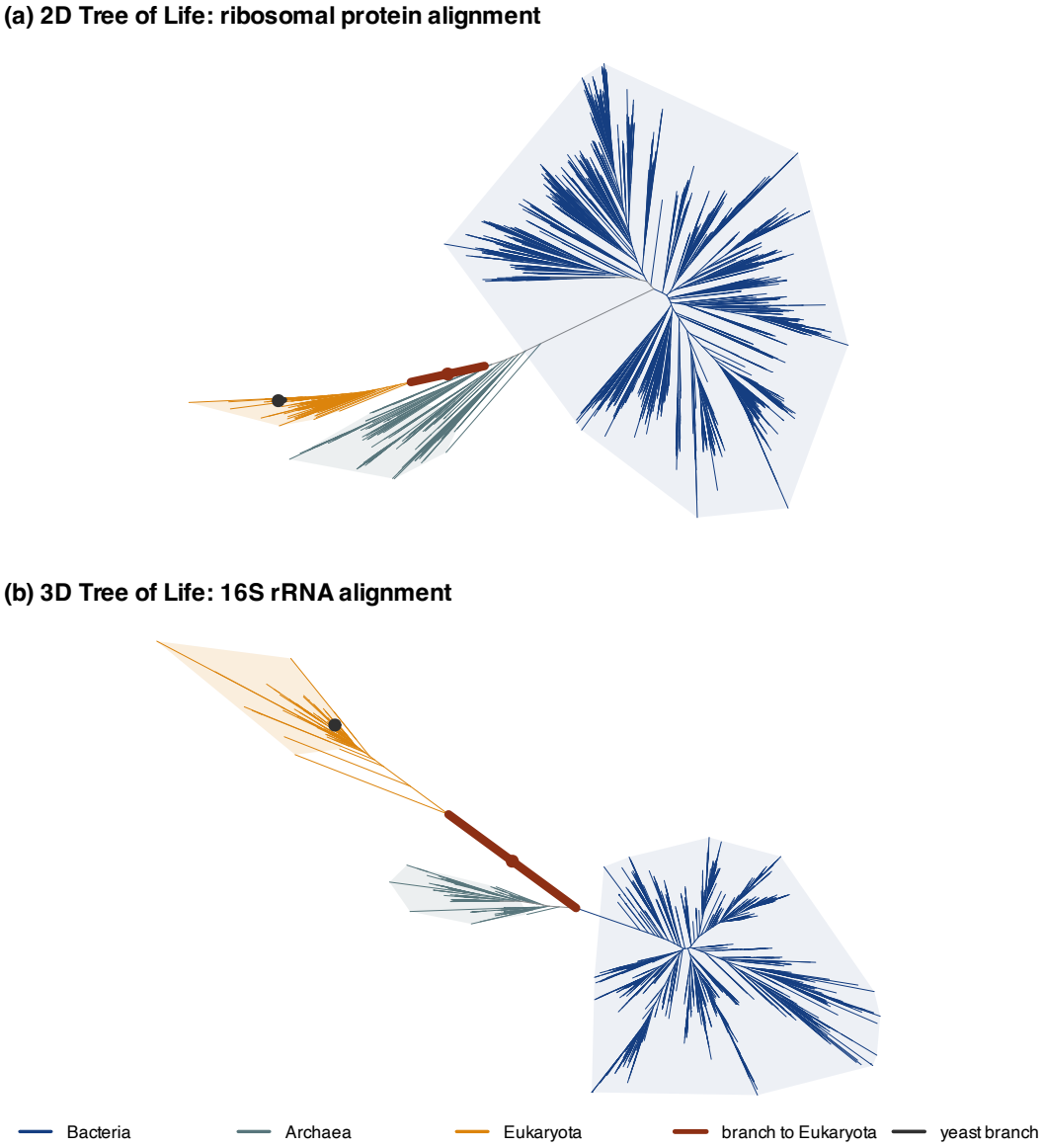



Figure S2: **Tree-of-Life reference trees. a,** Ribosomal-protein 2D tree. **b,** 16S rRNA 3D tree. Colors and hulls identify Bacteria, Archaea and Eukaryota. The highlighted internal segment marks the branch leading to Eukaryota; the black marker identifies the yeast branch.

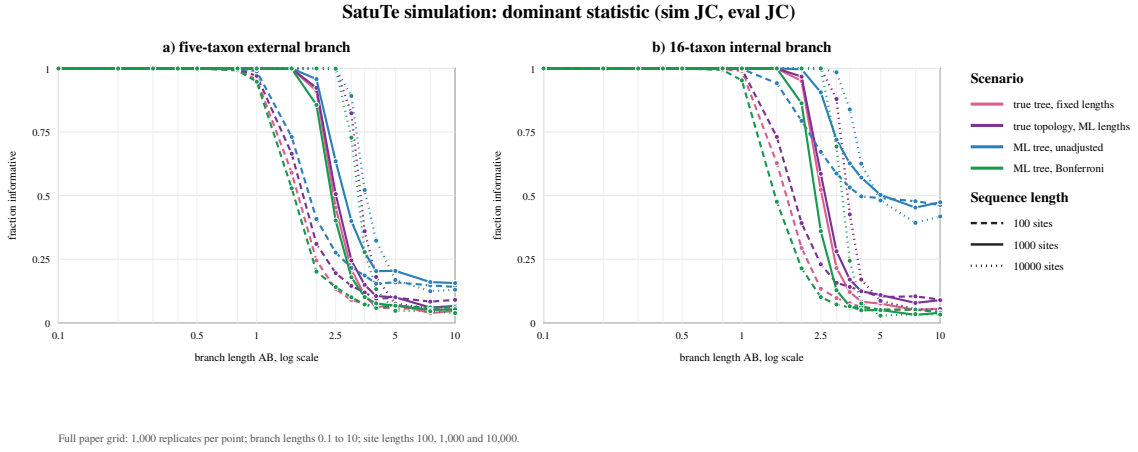



Figure S3: **Direct comparison with the original SatuTe simulation design.** The current implementation repeats the original paper's JC grid for five-taxon external and 16-taxon internal targets. Colors distinguish the true-tree reference, maximum-likelihood branch-length estimation, reuse of the alignment for topology selection and testing, and taxon-pair Bonferroni correction. Line styles show 100, 1,000 and 10,000 sites; every point contains 1,000 replicates.

| | Two external attachments | | | One external attachment | | | Correct topology (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| $t$ | 100 | 1,000 | 10,000 | 100 | 1,000 | 10,000 | 100 | 1,000 | 10,000 |
| 0.1 | 0 | 0 | 0 | 10 | 0 | 0 | 63.2 | 100.0 | 100.0 |
| 0.5 | 29 | 0 | 0 | 201 | 0 | 0 | 29.0 | 99.3 | 100.0 |
| 1.0 | 317 | 0 | 0 | 442 | 9 | 0 | 4.5 | 82.5 | 100.0 |
| 2.0 | 736 | 386 | 1 | 238 | 427 | 22 | 0.1 | 4.3 | 74.4 |
| 3.0 | 783 | 697 | 378 | 206 | 279 | 475 | 0.0 | 0.3 | 3.2 |
| 5.0 | 834 | 776 | 699 | 160 | 211 | 291 | 0.0 | 0.0 | 0.0 |
| 10.0 | 828 | 809 | 722 | 162 | 182 | 270 | 0.0 | 0.0 | 0.0 |

Table S2: **Representative split-placement results.** Counts among 1,000 inferred 16-taxon ML trees in which the branch separating the two eight-taxon subtrees attaches two external branches or one external and one internal branch, together with full-topology reconstruction accuracy. The omitted intermediate branch lengths follow the same monotone deterioration reported in the baseline experiment. Columns are alignment lengths in sites.

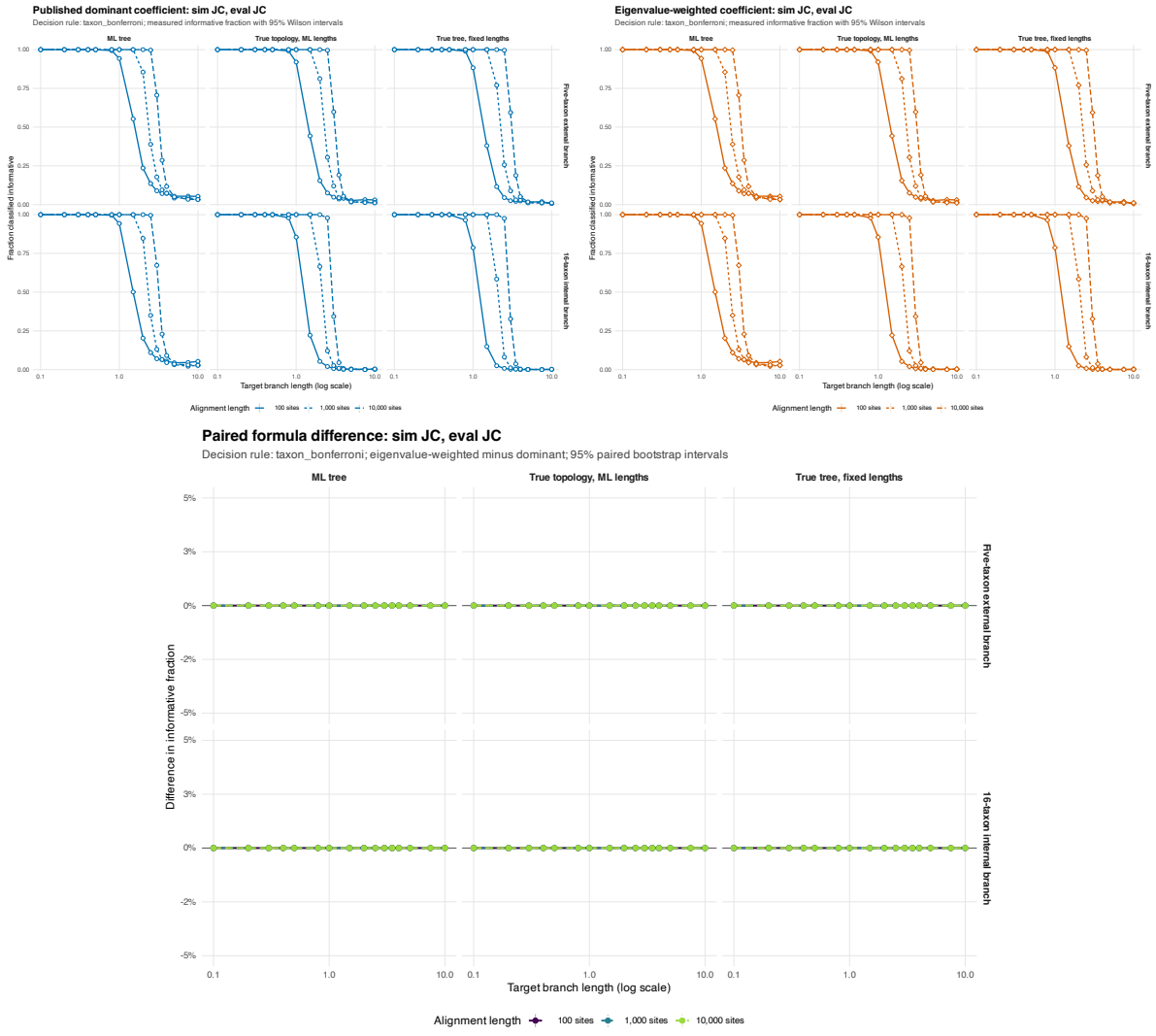



Figure S4: **JC power in the common comparison layout.** Dominant (top left), eigenvalue-weighted (top right) and paired difference (bottom) under taxon Bonferroni. The two formulas are identical under JC, so the paired difference is exactly zero in all 288 design cells.

| Focal $t$ | Split present | Mean bootstrap | Minimum bootstrap | SatuTe informative (%) |
|---|---|---|---|---|
| 0.1 | 993 | 93.32 | 14 | 100.0 |
| 0.2 | 999 | 99.19 | 74 | 100.0 |
| 0.5 | 1,000 | 99.99 | 97 | 100.0 |
| 0.8 | 1,000 | 100.00 | 100 | 99.1 |
| 1.0 | 1,000 | 100.00 | 100 | 92.4 |
| 1.5 | 1,000 | 100.00 | 100 | 49.7 |
| 2.0 | 1,000 | 100.00 | 100 | 21.6 |
| 2.5 | 1,000 | 100.00 | 100 | 9.6 |
| 3.0 | 1,000 | 100.00 | 100 | 5.9 |
| 3.5 | 1,000 | 100.00 | 100 | 4.3 |
| 4.0 | 1,000 | 100.00 | 100 | 4.6 |
| 5.0 | 1,000 | 100.00 | 100 | 3.8 |
| 7.5 | 1,000 | 100.00 | 100 | 3.5 |
| 10.0 | 1,000 | 100.00 | 100 | 4.3 |

Table S3: **Bootstrap–saturation contrast.** For 1,000 JC alignments of 100 sites on the 16-taxon tree, bootstrap support for the inferred split remains perfect at long focal branches while the Bonferroni SatuTe informative fraction approaches the 5% null level. This is the empirical basis for treating SatuTe as complementary to bootstrap support.

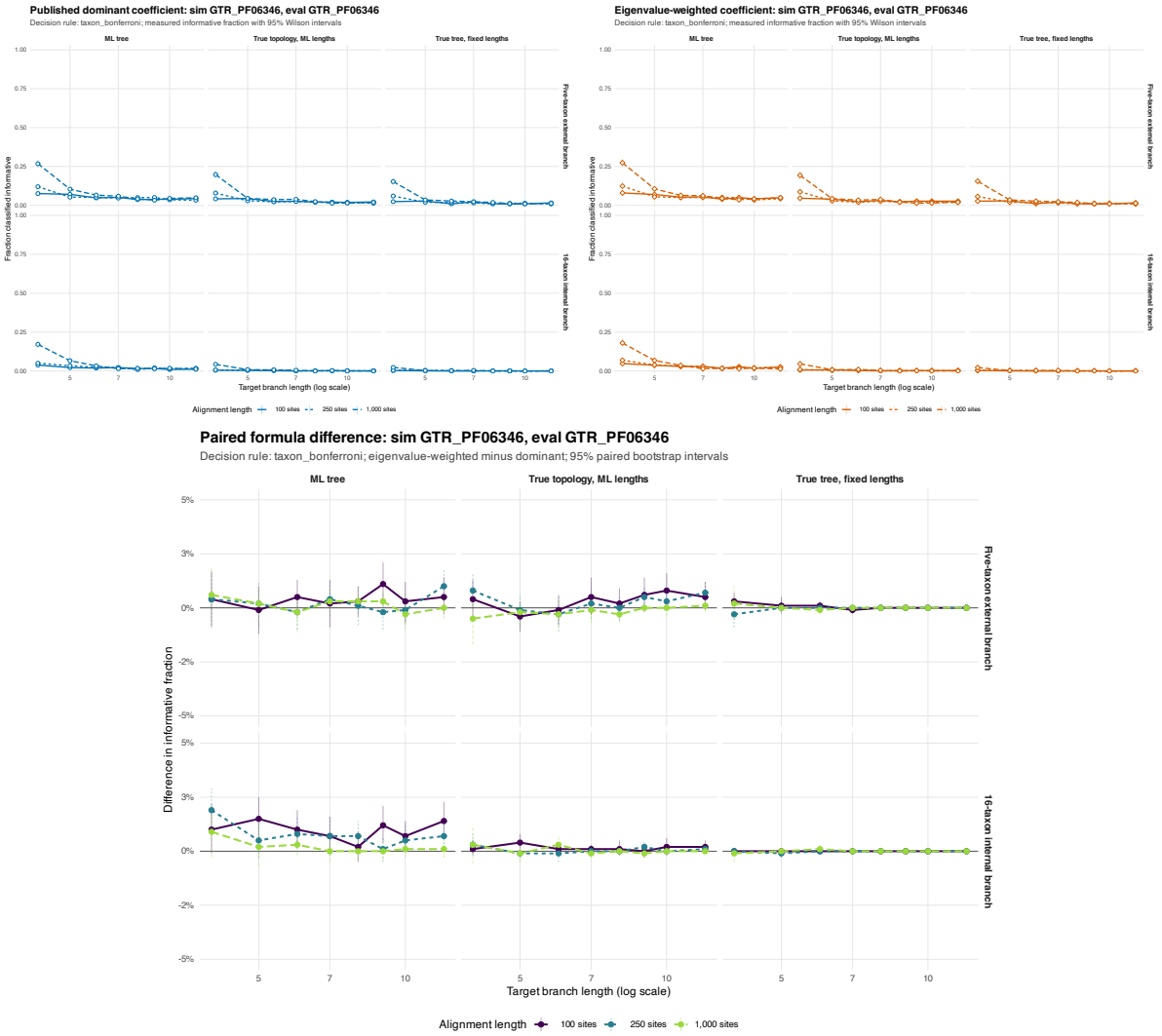



Figure S5: **Matched GTR+F power.** Dominant (top left), eigenvalue-weighted (top right) and paired difference (bottom) under taxon Bonferroni. Differences are small and mixed in this homogeneous-rate control.

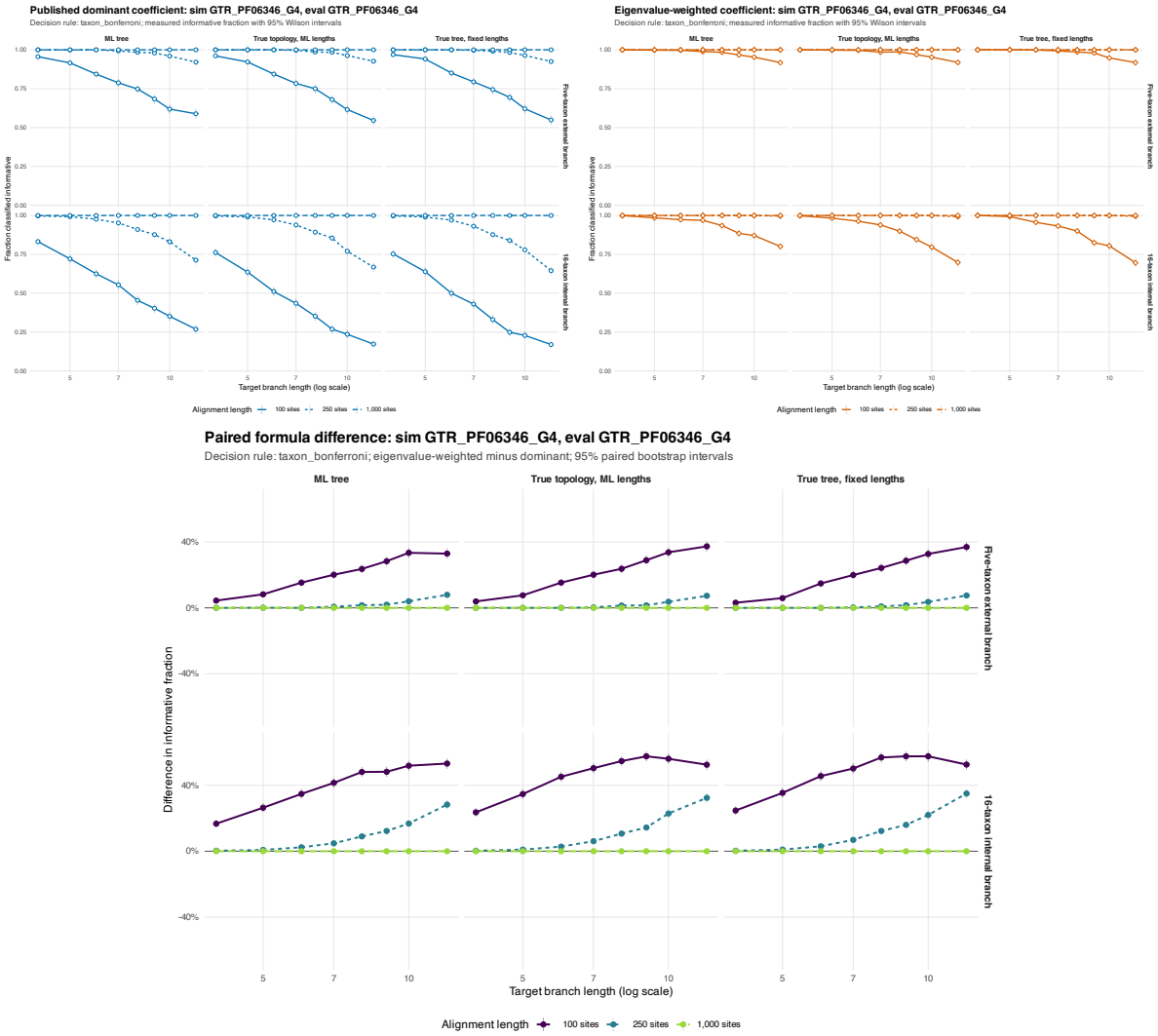



Figure S6: **Matched GTR+F+G4 power.** Dominant (top left), eigenvalue-weighted (top right) and paired difference (bottom) under taxon Bonferroni. The weighted formula is never worse across the evaluated cells and yields gains of up to 57.6 percentage points.

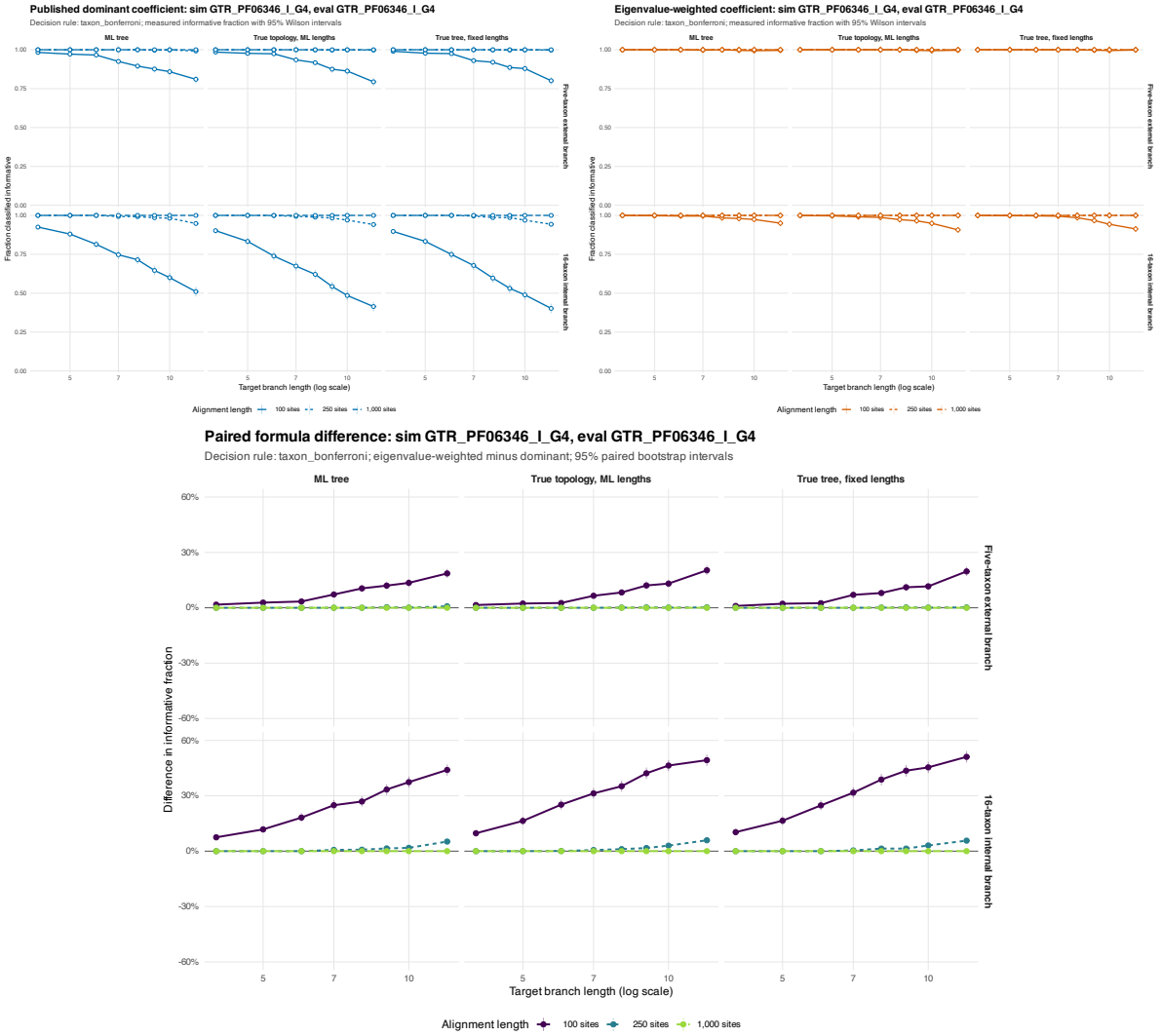



Figure S7: **Matched GTR+F+I+G4 power.** Dominant (top left), eigenvalue-weighted (top right) and paired difference (bottom) under taxon Bonferroni. The invariant category is handled by the corrected mixture null; weighting is never worse across the evaluated cells and gains up to 51.1 percentage points.

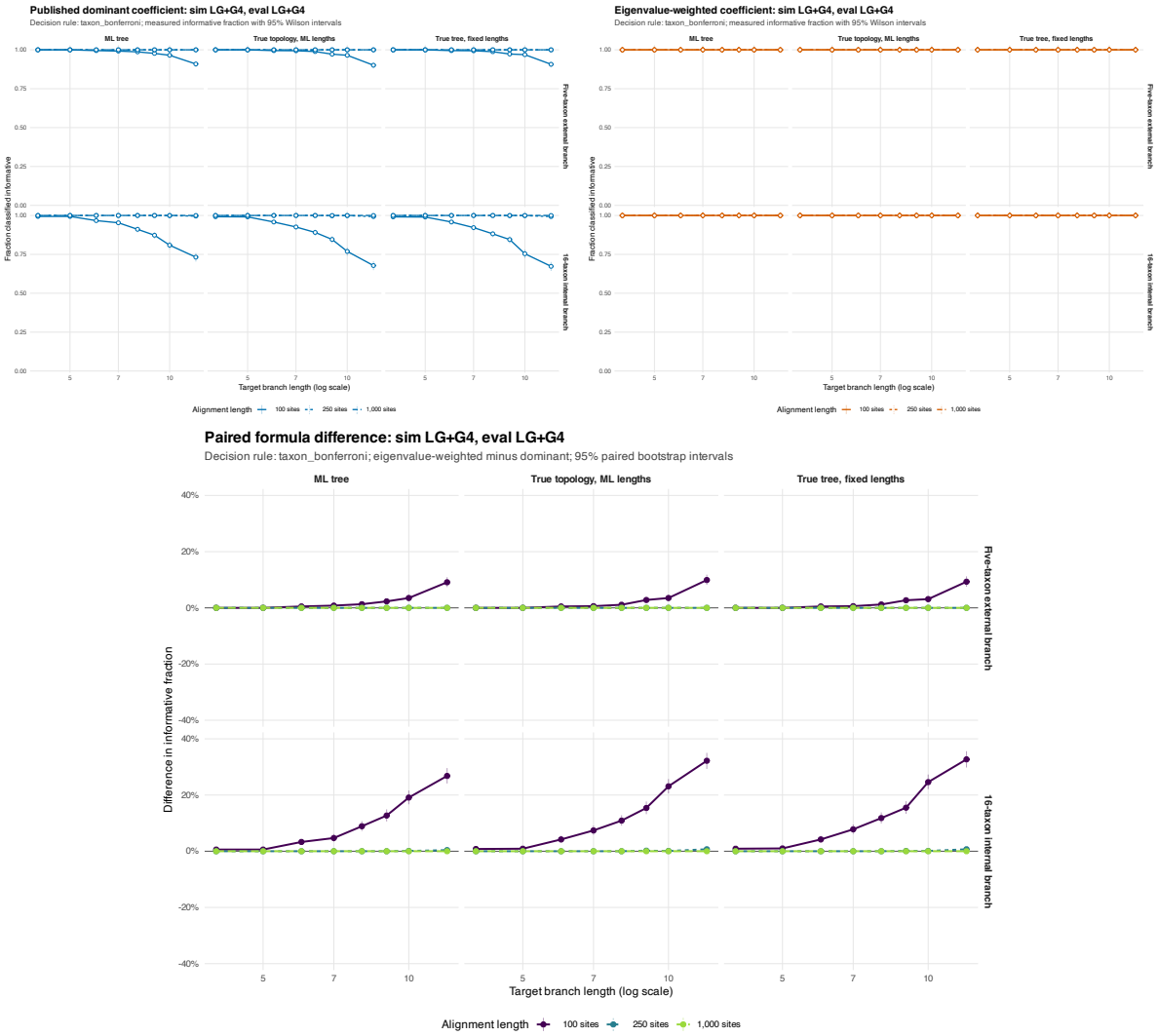



Figure S8: **Matched LG+G4 power.** Dominant (top left), weighted (top right) and weighted-minus-dominant difference (bottom) under taxon Bonferroni. Weighting is never worse across the evaluated cells and gains up to 32.7 percentage points.

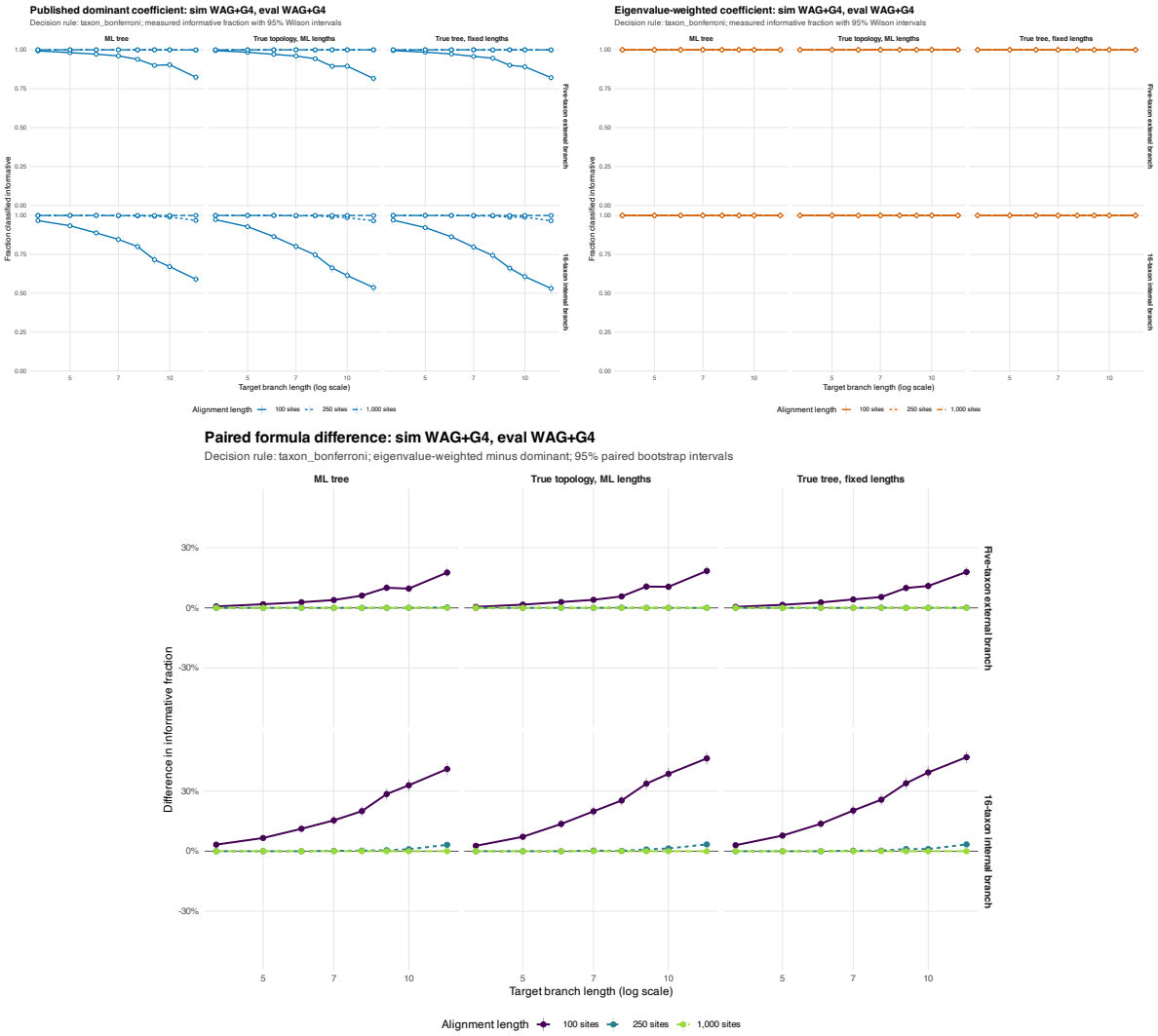



Figure S9: **Matched WAG+G4 power.** Dominant (top left), weighted (top right) and weighted-minus-dominant difference (bottom) under taxon Bonferroni. Weighting is never worse across the evaluated cells and gains up to 46.9 percentage points.

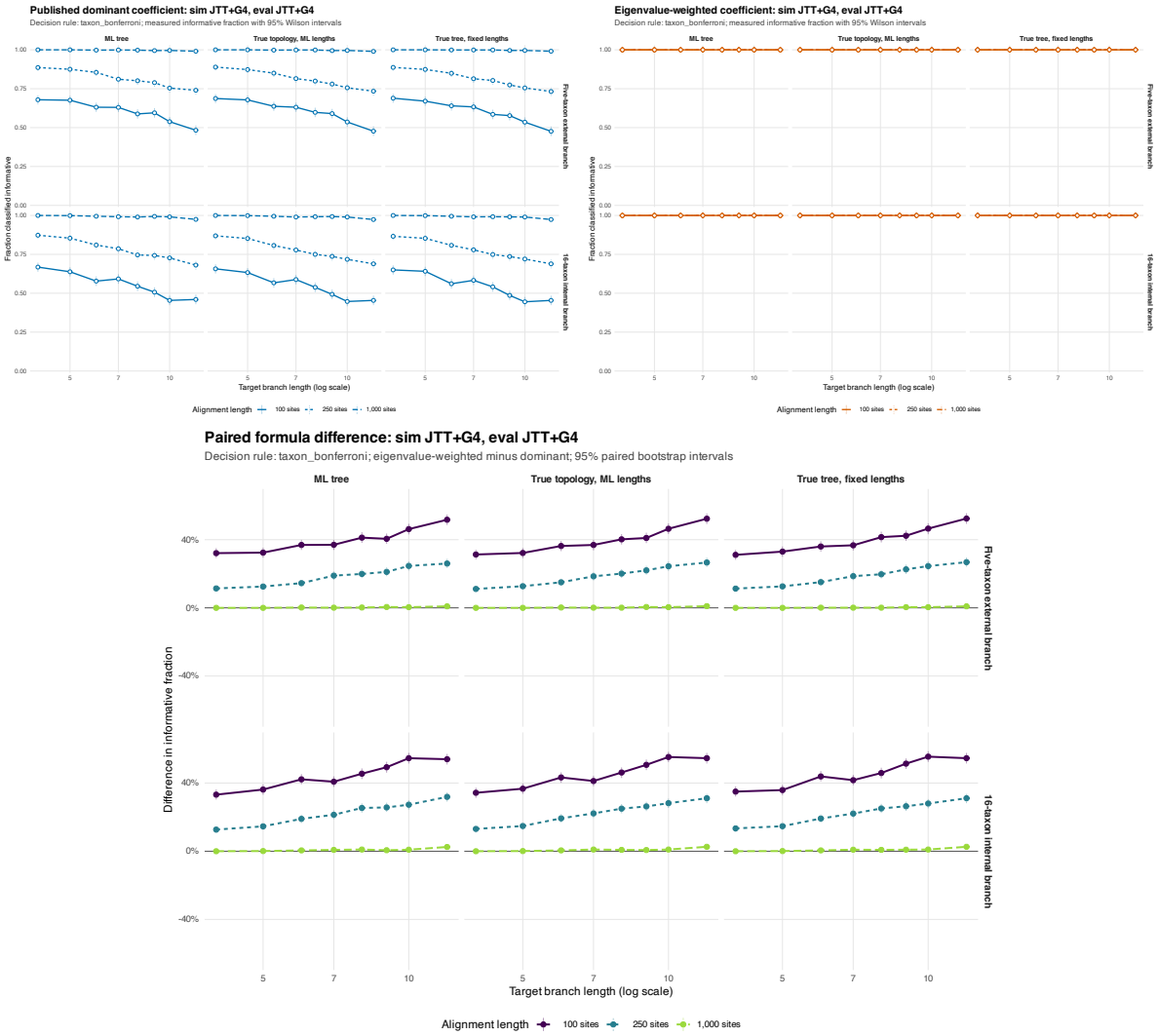



Figure S10: **Matched JTT+G4 power.** Dominant (top left), weighted (top right) and weighted-minus-dominant difference (bottom) under taxon Bonferroni. Weighting is never worse across the evaluated cells and gains up to 55.5 percentage points.

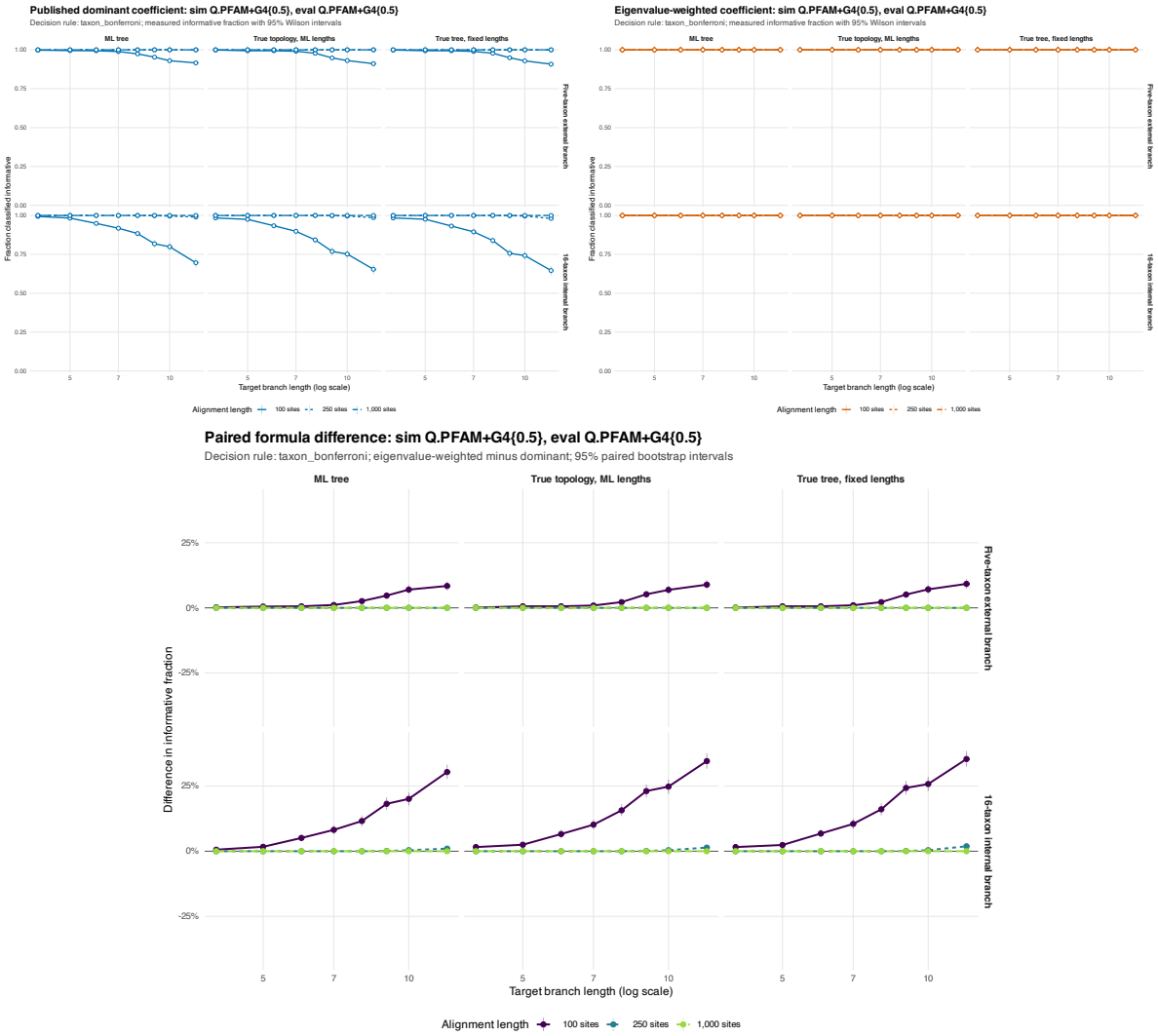



Figure S11: **Matched Q.pfam+G4 power.** Dominant (top left), weighted (top right) and weighted-minus-dominant difference (bottom) under taxon Bonferroni. Weighting is never worse across the evaluated cells and gains up to 35.4 percentage points.

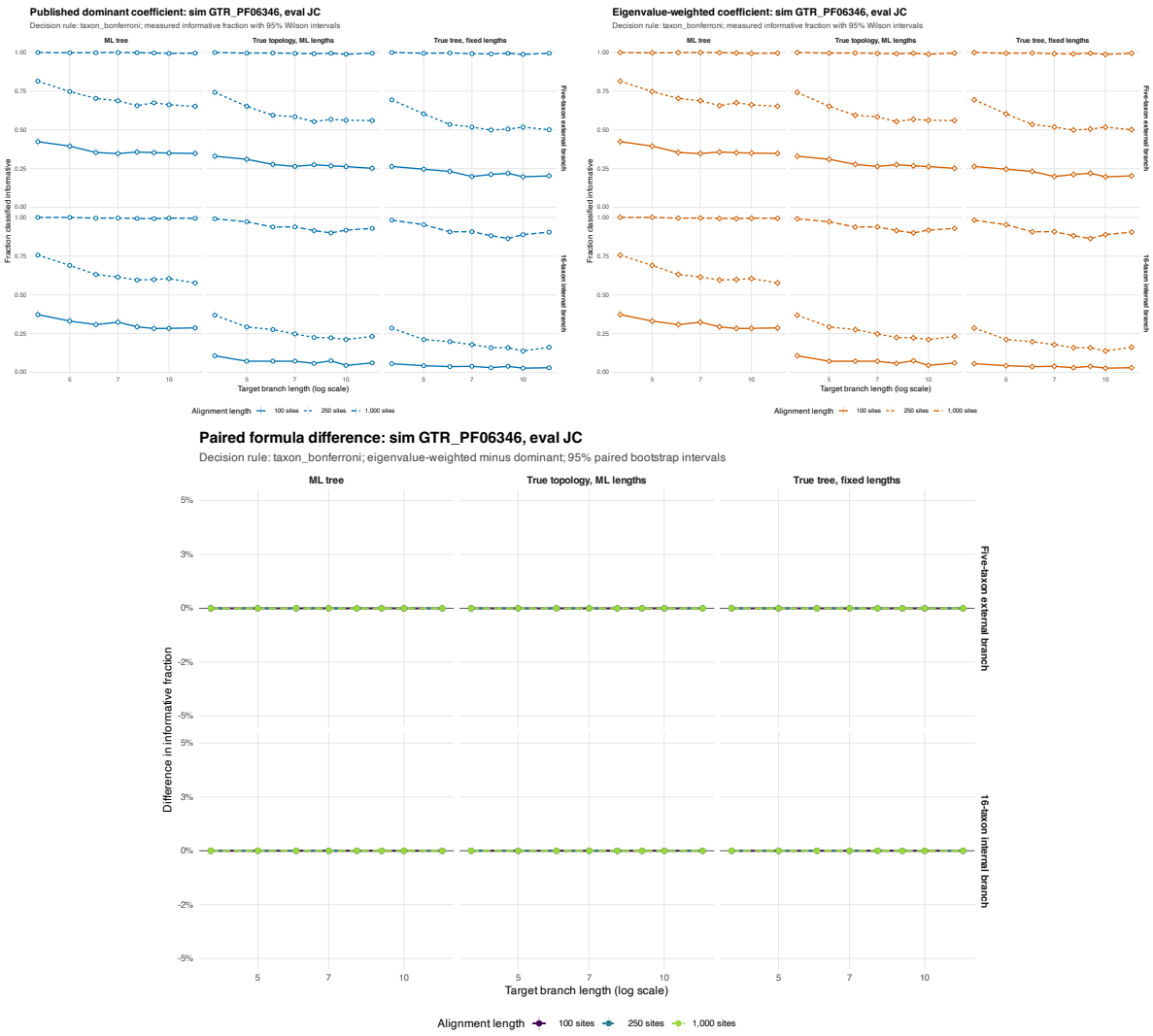



Figure S12: **GTR data evaluated by JC.** Dominant (top left), eigenvalue-weighted (top right) and paired difference (bottom) under taxon Bonferroni. The two formulas coincide under the JC evaluation model.

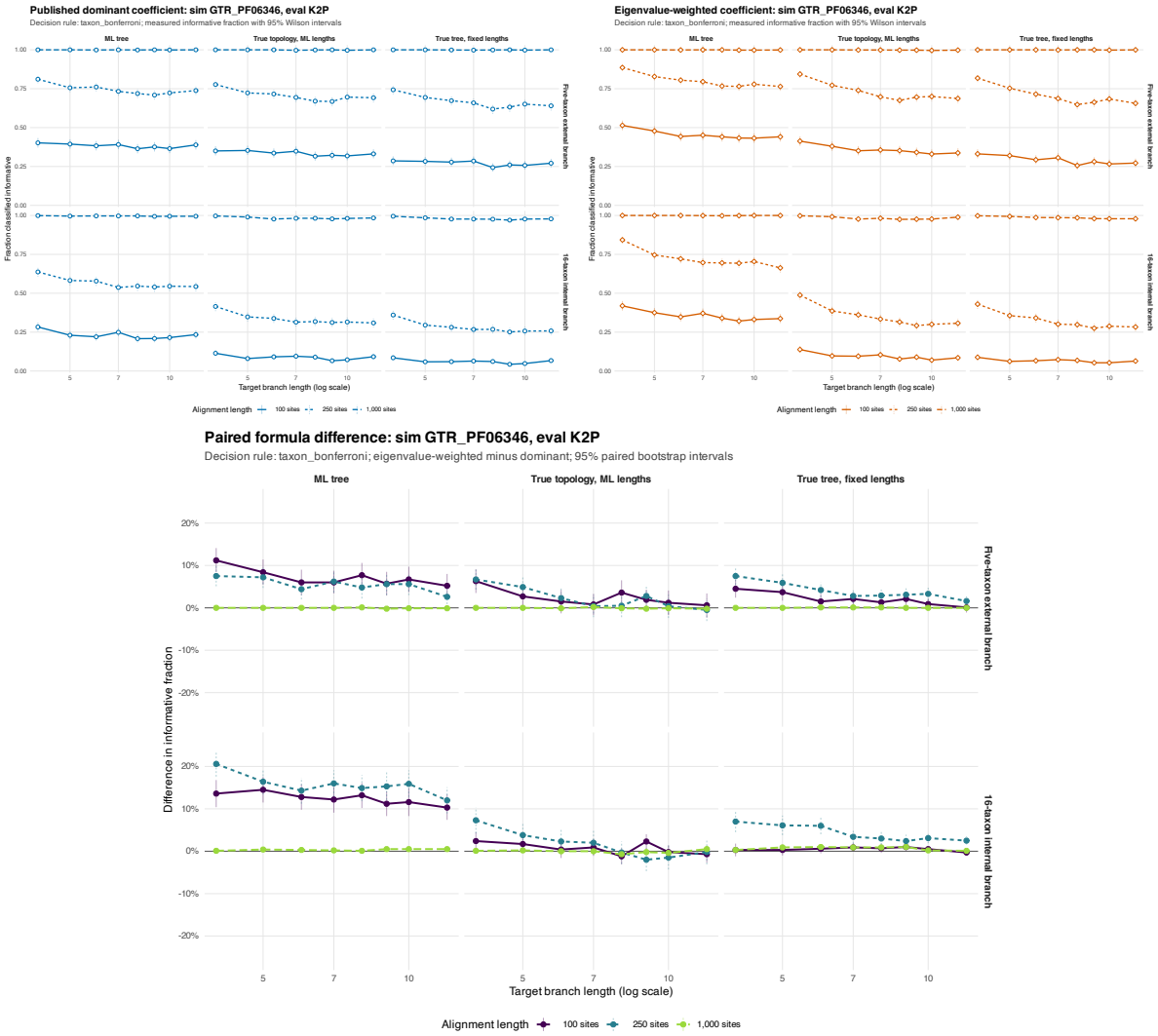



Figure S13: **GTR data evaluated by K2P.** Dominant (top left), eigenvalue-weighted (top right) and paired difference (bottom) under taxon Bonferroni. Most cells favour weighting, although a minority show small negative differences.

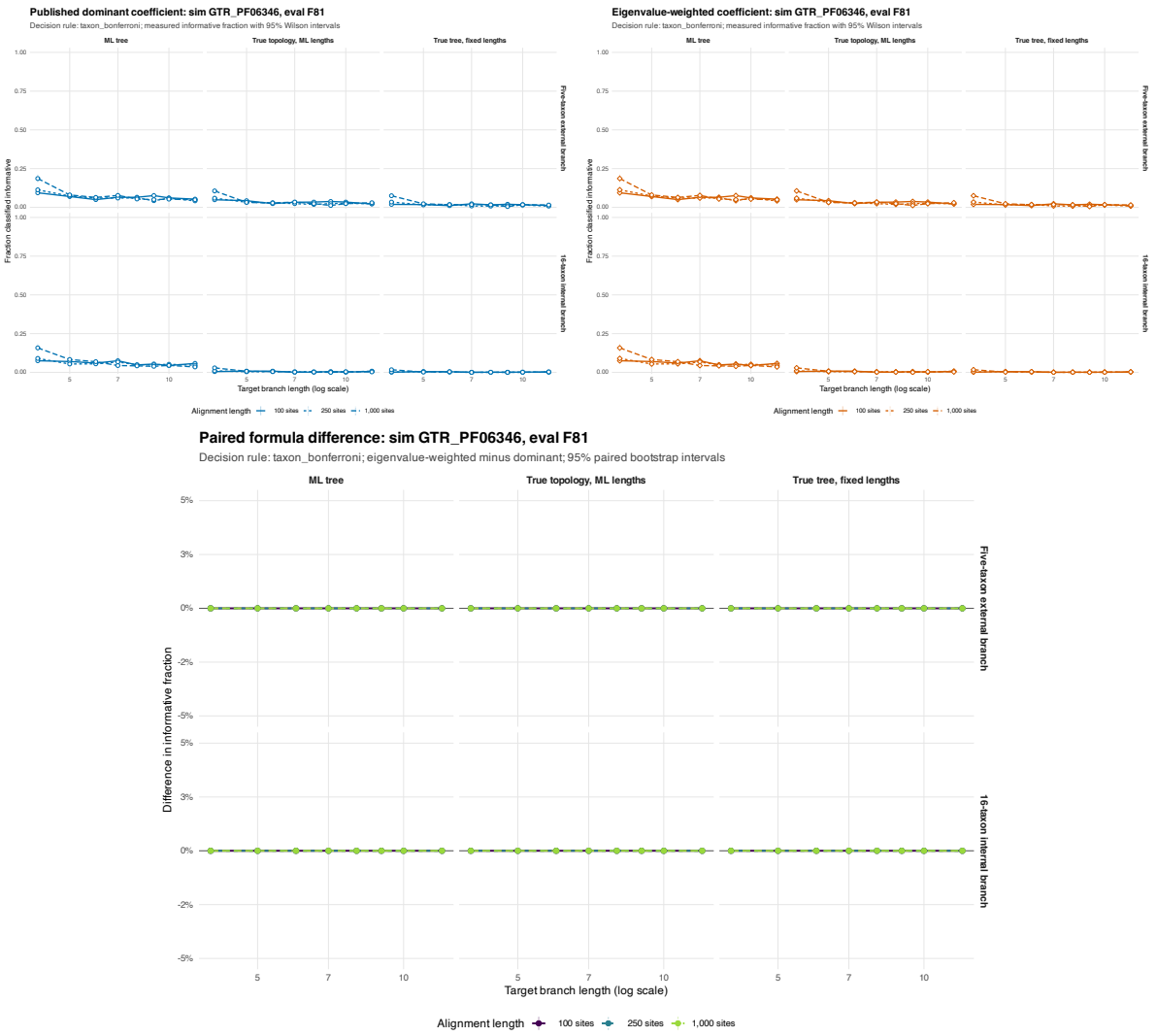



Figure S14: **GTR data evaluated by F81.** Dominant (top left), eigenvalue-weighted (top right) and paired difference (bottom) under taxon Bonferroni. The two formulas coincide under this evaluation model.

| Protein | Sites | 2D | 3D | $\Delta Z$ | Protein | Sites | 2D | 3D | $\Delta Z$ |
|---|---|---|---|---|---|---|---|---|---|
| L2 | 317 | 11.68 | 11.65 | 0.03 | L18 | 141 | 8.23 | 8.11 | 0.12 |
| L3 | 228 | 11.62 | 11.07 | 0.55 | L22 | 174 | 8.28 | 8.69 | -0.41 |
| L4 | 245 | 11.06 | 11.02 | 0.04 | L24 | 81 | 7.77 | 7.68 | 0.09 |
| L5 | 164 | 12.31 | 12.21 | 0.10 | S3 | 216 | 10.85 | 10.51 | 0.34 |
| L6 | 220 | 10.34 | 10.27 | 0.07 | S8 | 141 | 10.03 | 9.77 | 0.26 |
| L14 | 125 | 9.96 | 9.01 | 0.95 | S10 | 104 | 8.91 | 9.04 | -0.13 |
| L15 | 122 | 7.72 | 8.75 | -1.03 | S17 | 74 | 7.92 | 7.37 | 0.55 |
| L16 | 153 | 6.35 | 6.30 | 0.05 | S19 | 91 | 8.35 | 8.01 | 0.34 |

Table S4: **Per-protein topology comparison.** SatuTe $Z$-scores for the Eukaryota branch under the protein 2D tree and the rearranged 3D tree, with $\Delta Z = Z_{2D} - Z_{3D}$. All proteins remain significant under both placements; L15, L22 and S10 favour the rearranged 3D placement.

| rRNA region | Alignment sites | 3D $Z$ | Rearranged 2D $Z$ | $Z_{2D} - Z_{3D}$ |
|---|---|---|---|---|
| R1 | 0–114 | 4.57 | 5.08 | 0.51 |
| R2 | 115–431 | 6.13 | 5.94 | -0.19 |
| R3 | 432–621 | 3.63 | 3.45 | -0.17 |
| R4 | 622–1066 | 6.81 | 6.83 | 0.03 |
| R5 | 1067–1221 | 8.99 | 8.93 | -0.06 |
| R6 | 1222–1394 | 8.55 | 8.31 | -0.23 |
| R7 | 1395–1565 | 8.93 | 8.35 | -0.59 |
| R8 | 1566–1765 | 12.44 | 12.33 | -0.12 |
| R9 | 1766–1946 | 9.33 | 9.65 | 0.33 |

Table S5: **rRNA-region topology comparison.** Six of nine regions have negative $Z_{2D} - Z_{3D}$ and therefore a larger score under the 3D placement.

| Partition | Branch | Windows | Dominant | Weighted | Weighted fraction |
|---|---|---|---|---|---|
| Positions 1+2 | Genotype 1 | 507 | 1 | 1 | 0.2% |
| | Genotype 2 | 507 | 4 | 1 | 0.2% |
| | Genotype 3 | 507 | 5 | 1 | 0.2% |
| Position 3 | Genotype 1 | 252 | 11 | 6 | 2.4% |
| | Genotype 2 | 252 | 34 | 24 | 9.5% |
| | Genotype 3 | 252 | 33 | 27 | 10.7% |

Table S6: **HCV saturated-window counts.** Counts use $Z < z_{0.01}$ for three subtype-pair branches. Positions 1+2 and position 3 contain equal 36-site windows but span different genomic widths.

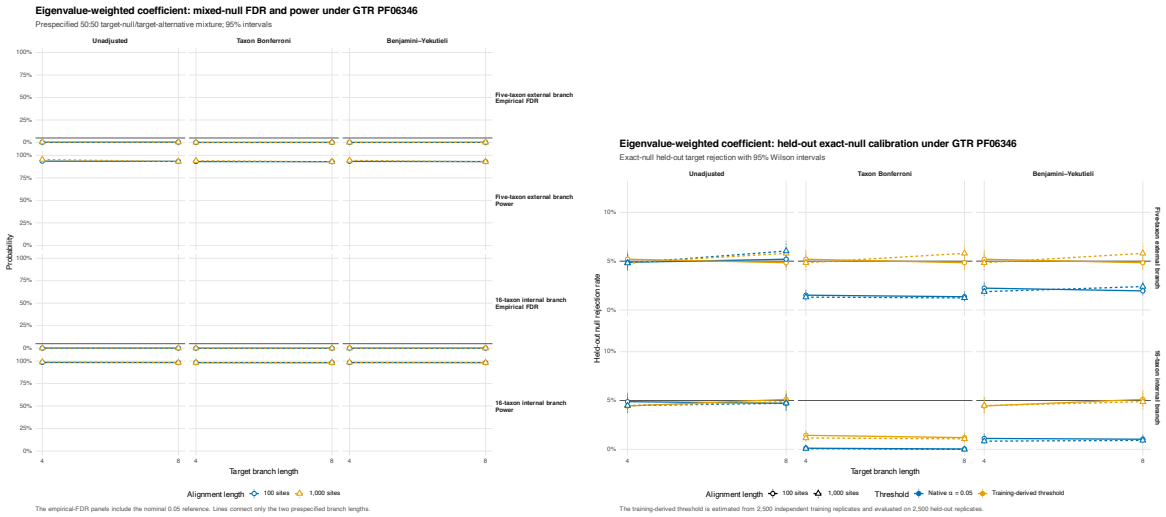



Figure S15: **Truth-labelled GTR calibration. a,** Empirical FDR and power of the weighted formula in a prespecified 50:50 mixed-null design. **b,** Native and training-derived rejection on 2,500 held-out exact nulls per cell. Columns show unadjusted, taxon-Bonferroni and BY rules.